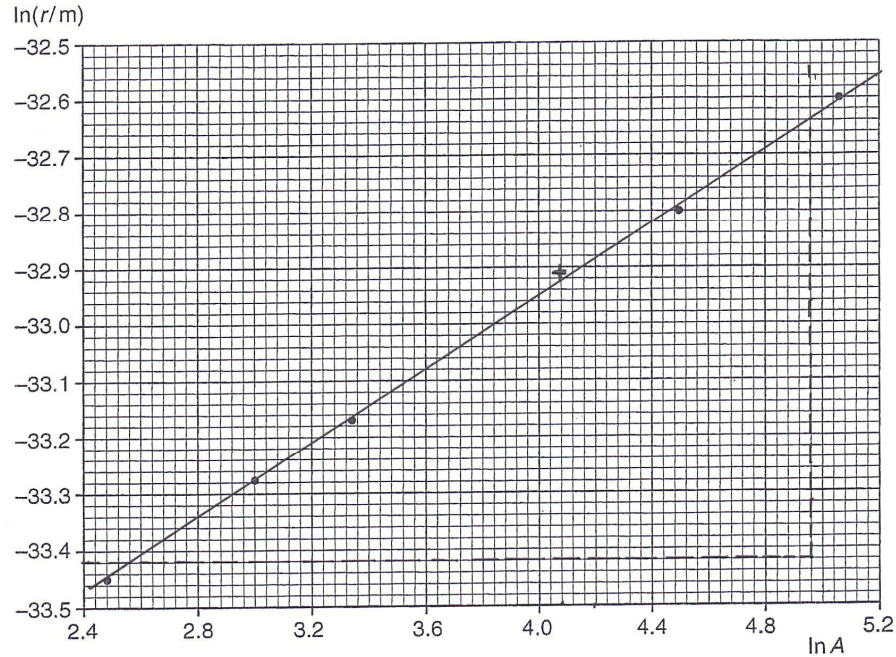


6 (a) (i) $A=60$
from the graph, $r = 5.1 \times 10^{-15} \text{ m}$
 $\ln r = -32.91$

(ii) $\ln A = 4.094$, $\ln r = -32.91$



(iii) The two points on the best fit line are (2.56, -33.42) and (4.96, -32.64)

$$\text{gradient} = \frac{(-33.42) - (-32.64)}{2.56 - 4.96} = 0.325$$

(iv) equation of line on Fig. 6.2

$$\ln r = \ln k + n \ln A$$

$$-33.42 = \ln k + (0.325)(2.56)$$

$$\ln k = -34.252$$

$$k = 1.33 \times 10^{-15} \text{ m}$$

(v) The value of k represents the radius of a nucleon (can be a proton or a neutron).

(b) (i) radius of U-235 nucleus = kA^n

$$= 1.33 \times 10^{-15} (235)^{0.325}$$

$$= 7.85 \times 10^{-15} \text{ m}$$

$$\text{(ii) density of H-1 nucleus} = \frac{1u}{\frac{4}{3}\pi r^3} = \frac{1u}{\frac{4}{3}\pi (k1^{0.325})^3} = \frac{1u}{\frac{4}{3}\pi (k)^3}$$

A=1 for H-1 nucleus

$$\text{density of U-235 nucleus} = \frac{235u}{\frac{4}{3}\pi r^3} = \frac{235u}{\frac{4}{3}\pi (k235^{0.325})^3}$$

$$= \frac{235u}{\frac{4}{3}\pi (k^3 235^{0.975})}$$

A=235 for U-235 nucleus

$$\text{ratio} = \frac{1u}{\frac{4}{3}\pi(k)^3} \times \frac{\frac{4}{3}\pi(k^3 235^{0.975})}{235u} = \frac{(235^{0.975})}{235} = 235^{-0.025} = 0.872$$

- (c)
1. Answer in b(ii) compares the density of nucleus but in c(i) it is comparing two different states. One being a gas and the other a solid. Even when comparing the same substance at gaseous and solid phase, they will be different. Solid has a much higher and tighter packing compared to gas.
 2. The given ratio compares the average density of many atoms whereas the ratio in b(ii) compares the density of single nuclei.